

Chapter 1

Introduction

The petrochemical industry in Bangladesh is growing, fueled by industrial demand and natural gas resources. Local ventures like Super Petrochemical Limited (SPL) now produce key chemicals such as hexane and benzene, reducing import reliance and supporting national industrial growth.

1.1 BACKGROUND

Super Petrochemical Limited (SPL) stands as the first petrochemical industry in the private sector of the country and has been a pioneer in its field since its establishment in July 2013. The company was founded with the vision of reducing the nation's dependency on imported petroleum and petrochemical products by utilizing both locally available and imported raw materials.

SPL plays a vital role in supporting the national economy through the production of import-substitute products derived from feedstocks such as Natural Gas Condensate and Naphtha. The company's operations are structured around three primary production units: the Naphtha Production Facility, Naphtha Processing Facility, and the Hexane Production Facility, the latter of which is responsible for the production of key petrochemicals like Hexane and Benzene.

In addition to its core production units, SPL is supported by several Auxiliary and Utility Units that ensure smooth, efficient, and uninterrupted industrial operations. With its advanced processing systems and commitment to industrial innovation, SPL continues to contribute significantly to the country's industrial development and energy sector.



Figure 1.1 Logo of SPL

1.2 COMPANY PROFILE



Figure 1.2 : Super PetroChemical Ltd.(SPL)

Name of the factory	: Super Petrochemical LTD.(SPL)
Factory	: Dangarchar, Juldha
Location	: PO- Charlakkha, Thana: Karnaphuli Chittagong-4371, Bangladesh
Year of Established	: July, 2013
Processing Capacity	: Naphtha Production Facility: 6700 Barrels Per Day (BPD) • Condensate Splitter Unit: 2700 BPD • Condensate Fractionation Unit: 4000 BPD • Designed By: Beijing Zehua Chemical Engineering Co Ltd,China

Naphtha Processing Facility, Unit-1: 4000 BPD

- Designed By: Chemex LLC, USA.
- Catalyst Supplier: Axens, France.

Hexane Production Facility: 1000 BPD

- Designed By: AMT International, Inc. (AMT), USA.

Tank Terminal Storage Capacity : 79600 M3 (Total 27 Tanks)

- Raw Material Storage: 53100 M3
- Intermediate Storage: 1500 M3
- Finished Product Storage: 25000 M3

Products : 1. Gasoline
2. Diesel
3. MTT (Mineral Turpentine)
4. Mixed Xylene
5. Toluene
6. Hexane
7. HydroCarbon solvent
8. LPG

Customer : Eastern Refinery Limited, Bangladesh

Certification : ASTM, UOP, IP, BDS

1.3 MISSION & VISION

Mission:

- Sustain Leadership in energy conservation, efficiency, productivity, and innovation.
- Capitalize on emerging opportunities in the domestic market.
- Maintain global standards in health, safety, and environmental norms with a strong commitment towards community welfare.
- Continuing focus on employee welfare and employee relations.
- Imbibe the highest standards of business ethics and values.
- Primary responsibility is to develop and add value to the national resource.
- Objective is to contribute to the well-being of the people and the nation

Vision:

To be the market leader amongst the Refinery and Petrochemical Companies in Bangladesh, with a strong emphasis on Productivity, Customer Satisfaction, Safety, Health and Environment Management, Corporate Social Responsibility and Care for Employees & Shareholders.

Chapter 2

Process Flow & Overview of Plant

2.1 OVRALL PROCESS FLOW

SPL is in an agreement with RPGCL for the purchase of the raw material for Naphtha Production Facility. It also has an agreement with BPC to procure Raw Naphtha and correspondingly in agreement to sell Gasoline, Diesel, and Hydrocarbon solvents to subsidiaries of BPC for marketing and distribution. Since converting raw materials into finished products is a significant level of complex technology, we are in collaboration with the world's renowned technical experts in this field.

The operation process at Super Petrochemical Limited (SPL) follows a systematic and continuous flow involving several stages of chemical processing, heat exchange, cooling, and separation to produce refined petrochemical products. Below is a step-by-step description of the process:

1. Tank Farm

The process begins at the Tank Farm, where raw materials (feedstock such as naphtha or condensate) are stored before processing.

2. Feed Vessel

The feedstock is then transferred to the Feed Vessel, which regulates and stabilizes the feed before it enters the processing system.

3. Heat Exchanger & Heater

The feed is preheated using a Heat Exchanger and then further heated in a Heater to reach the desired temperature suitable for catalytic reactions.

4. Catalytic Reactor

The preheated feed enters the Catalytic Reactor, where chemical reactions occur in the presence of catalysts to reform and crack the hydrocarbon molecules.

5. Heat Exchanger (Cooling Stage)

After reaction, the product stream is cooled using another Heat Exchanger to bring down the temperature for further processing.

6. Aero Cooler & Accumulator Drum

Further cooling is achieved via an Aero Cooler, and the mixture is collected in an Accumulator Drum, which helps separate gaseous and liquid phases.

7. Product Collection

One stream from the Accumulator Drum leads directly to Product collection.

8. Pump for Reforming & Tower

Another stream is directed through a Pump for Reforming, pushing it into a Tower where separation and fractionation of components occur.

9. Aero Cooler, Drum, and Exchanger

The separated stream from the Tower is again cooled using an Aero Cooler, then passed into a Drum, and finally through an Exchanger for final temperature adjustment.

10. Final Product

After the final stage of conditioning, the processed streams are collected as finished products ready for storage or distribution.

Process Flow Diagram:

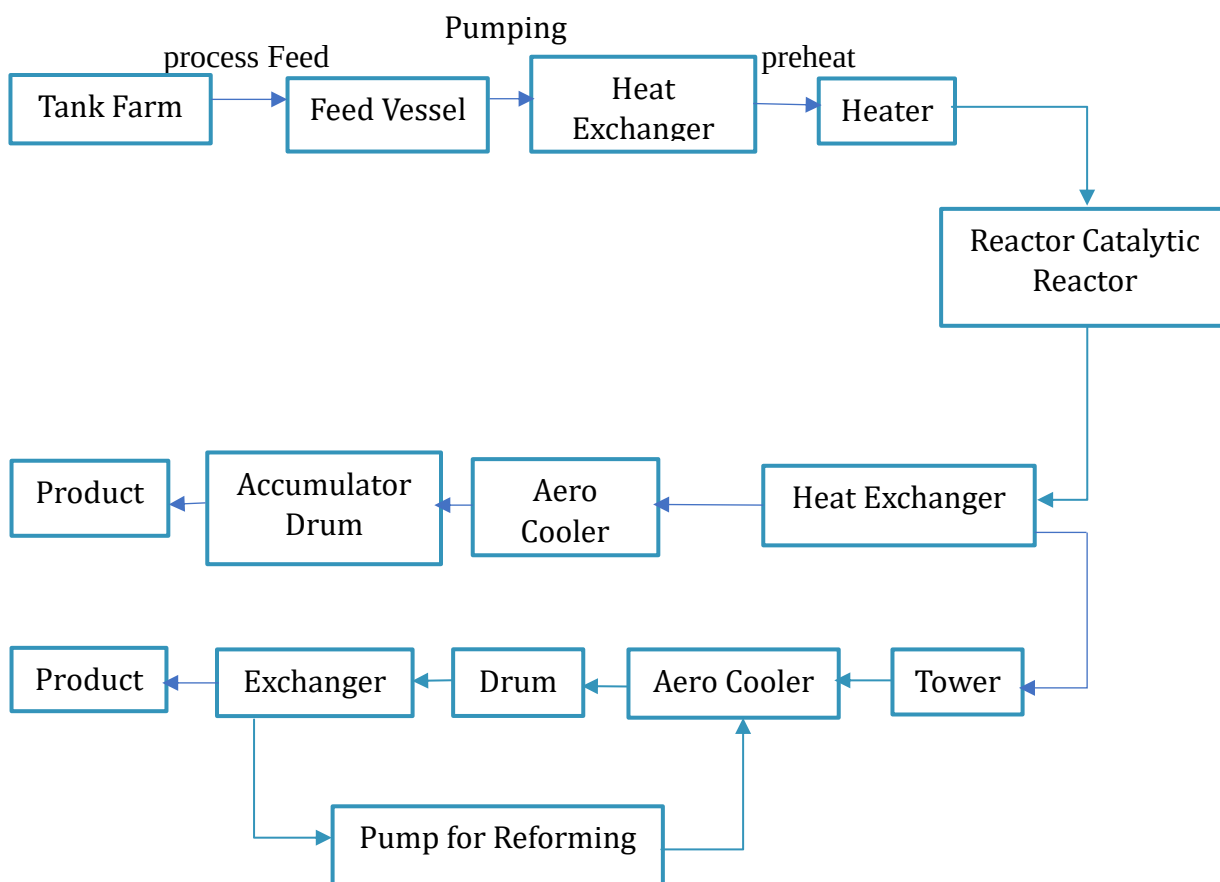


Figure 2.1: Process Flow Diagram of SPL

This process flow demonstrates SPL's integrated and energy-efficient approach to refining petrochemical feedstock into valuable industrial chemicals like hexane, benzene, and others. Each stage ensures quality control, safety, and maximum recovery of desired products.

2.2 MECHANICAL OVERVIEW

The mechanical department at Super Petrochemical Limited (SPL) plays a crucial role in ensuring smooth operations across the plant. Its responsibilities are categorized into three main areas:

1. Development Works

This includes planning and executing projects aimed at enhancing plant infrastructure and capacity. A major part of development Erection Works, which includes:

- Procurement of necessary mechanical components and materials.
- Construction/Fabrication, which is further divided into:
 - Fittings
 - Welding
 - Installation

2. Workshop Operation

The mechanical workshop handles repair, fabrication, and maintenance tasks necessary for day-to-day operations, supporting both development and ongoing plant activities.

3. Running Plant Operation Support

This section ensures the continuous functioning of the plant through:

- Static Maintenance – maintenance of non-moving equipment such as tanks, pipelines, and structural components, involving fabrication-related tasks like:
 - Fittings
 - Welding
 - Installation
- Rotating Maintenance – maintenance of moving equipment such as pumps, compressors, and motors, which are vital to the dynamic operations of the plant.

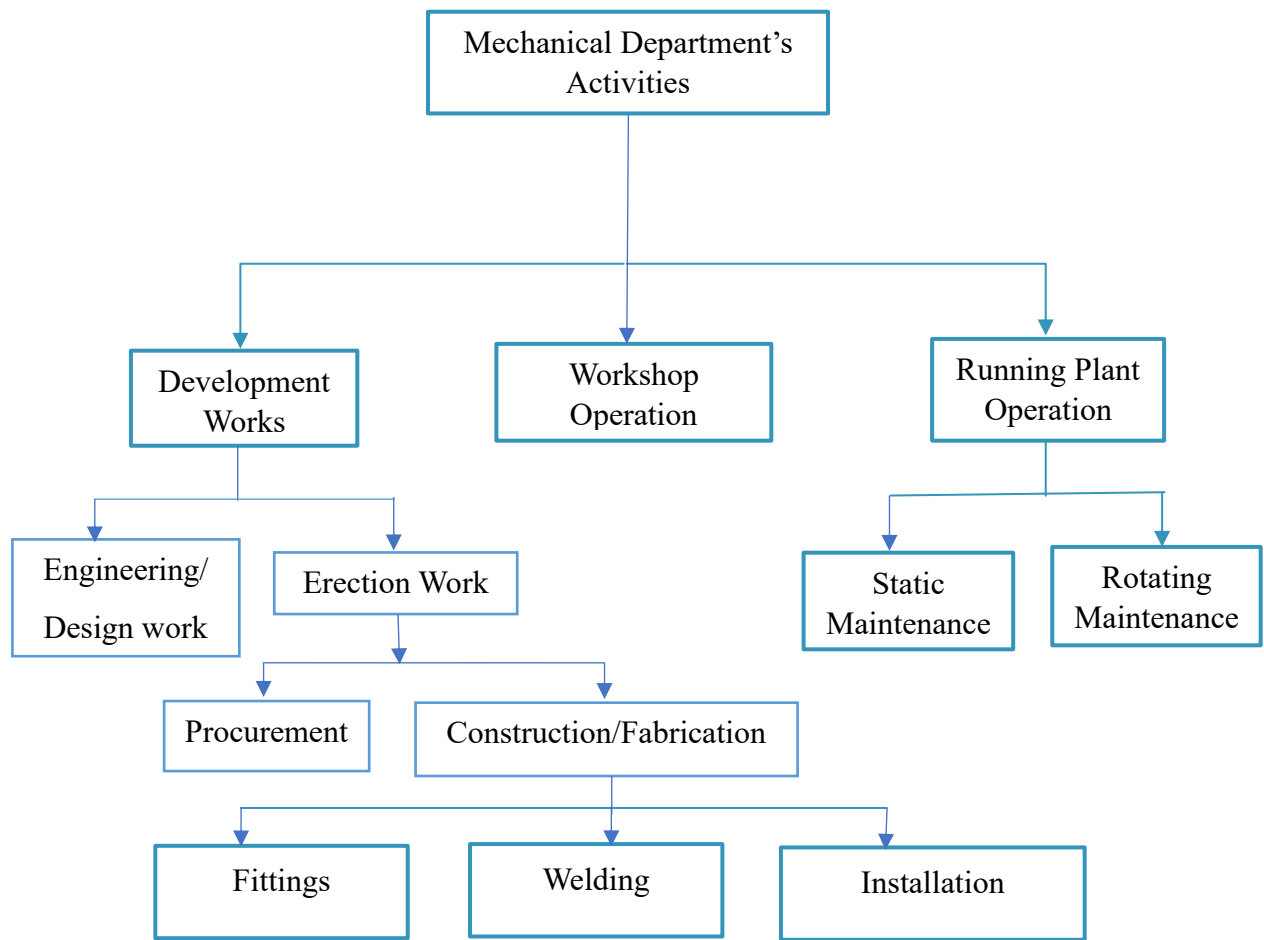


Figure 2.2: Mechanical Overview of SPL

2.3 OVERVIEW OF DIFFERENT SECTION

1. Rotating Equipment

- Pump
- Compressor
- Aero Cooler
- Cooling Tower
- Agitator

- **Cooling Tower:** A cooling tower is a device that removes heat from water by evaporating a small portion of it and releasing it into the atmosphere.

In Super Petrochemical Ltd. (SPL), a cooling tower serves as a critical component for managing heat generated during various industrial processes. Its primary function is to remove excess heat from circulating water used in heat exchangers, condensers, and reactors. As chemical reactions and mechanical operations produce significant thermal energy, the cooling tower helps maintain optimal temperatures by cooling the heated water and recirculating it throughout the system. This not only ensures the stable and efficient operation of key equipment such as pumps, compressors, and reactors, but also reduces the risk of overheating, which can lead to equipment failure or safety hazards. Additionally, by allowing for the reuse of cooled water, cooling towers contribute to water conservation and improved energy efficiency. They also play a role in meeting environmental regulations by minimizing thermal pollution. Overall, the cooling tower is essential for sustaining smooth, safe, and cost-effective operations in a petrochemical plant. In SPL the types of cooling tower are induced.

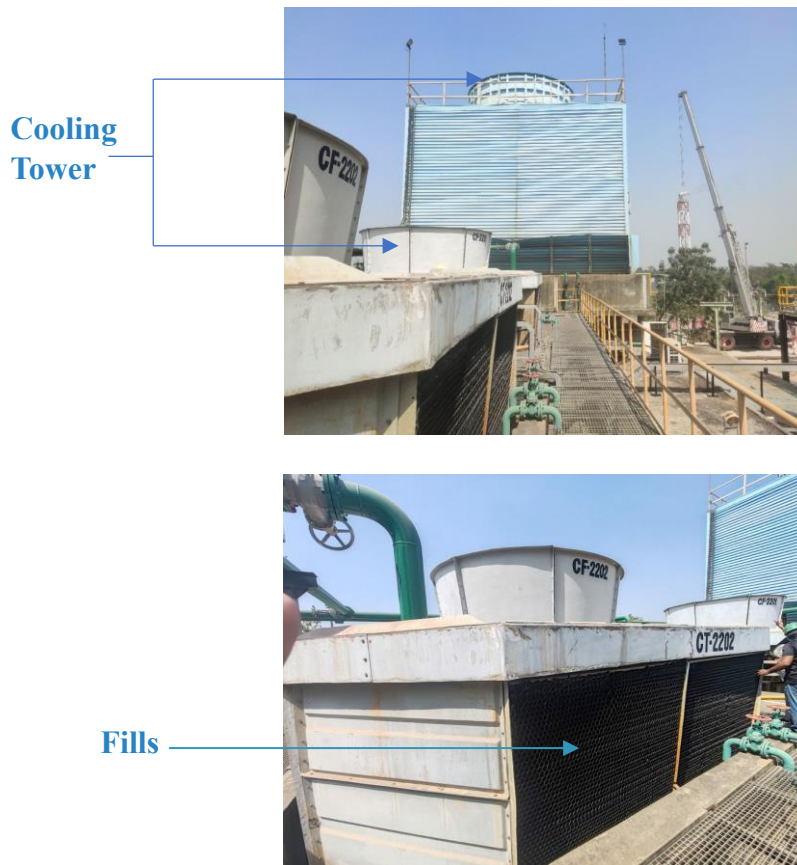


Figure 2.3: Cooling Tower

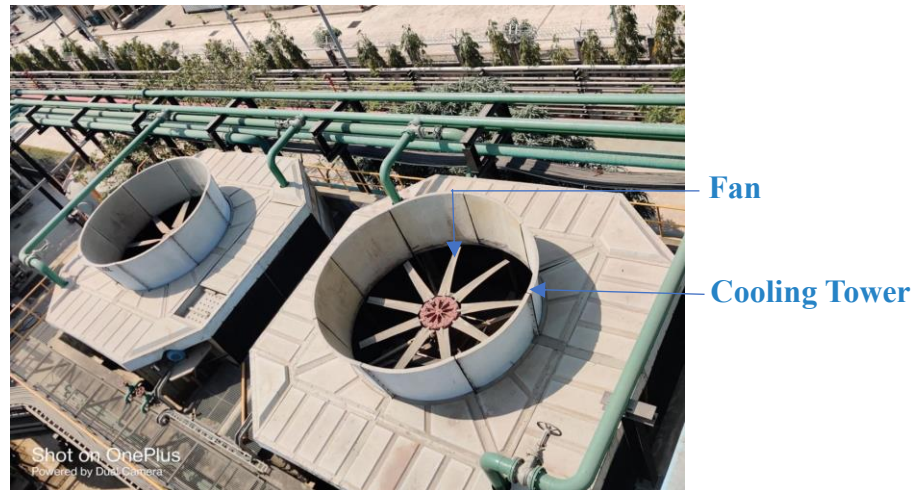


Figure 2.4: Fan and blade of Cooling Tower (Blade Angle: 5-7 degree)

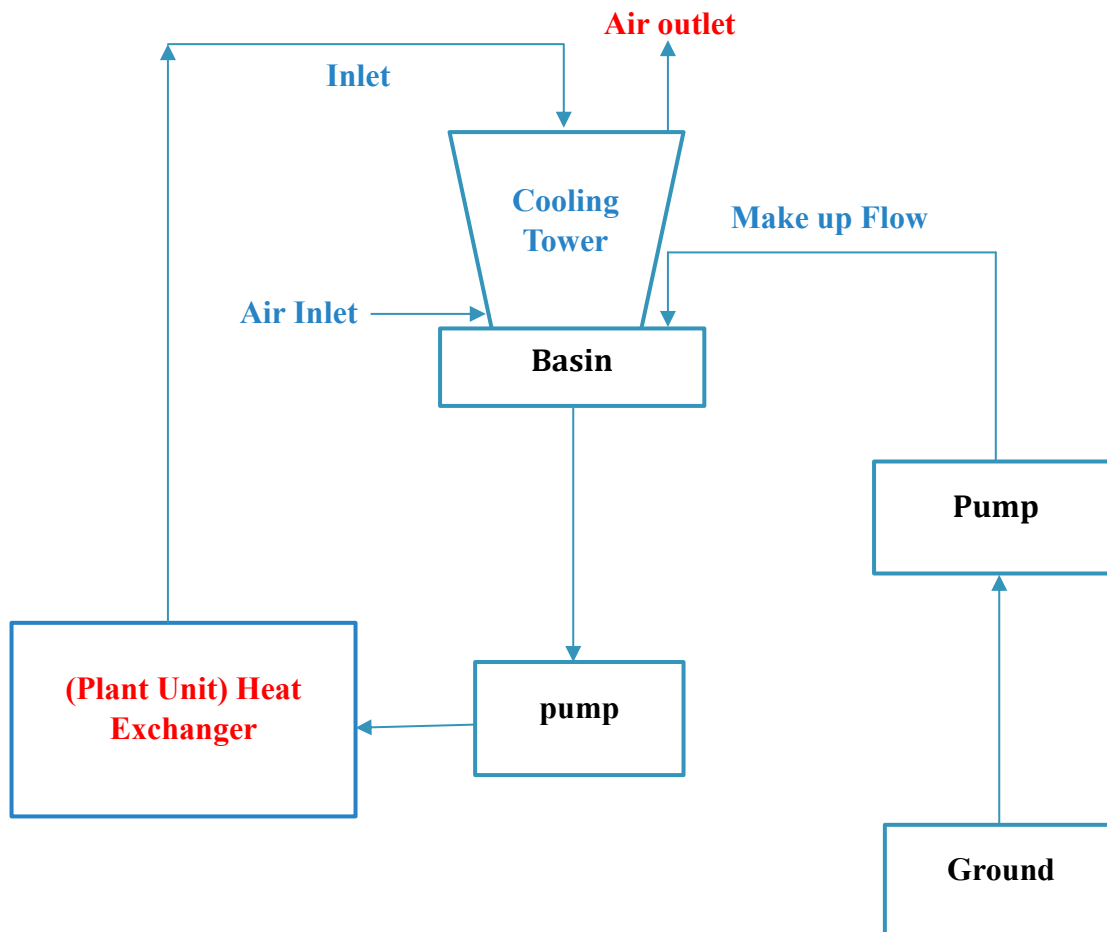


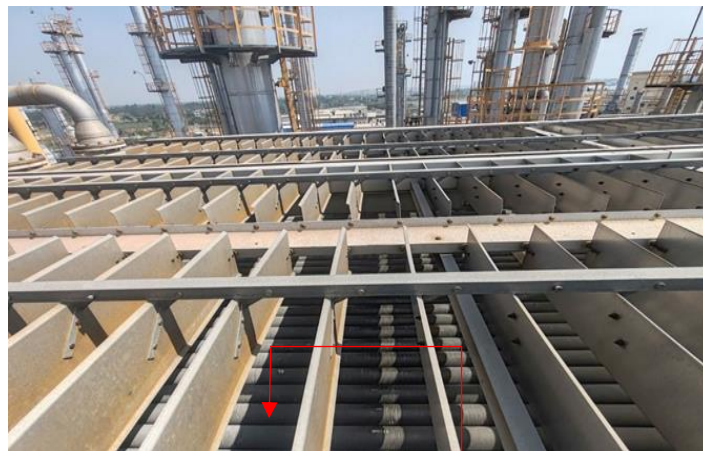
Figure 2.5: Flow chart of Cooling Tower in SPL

- **Aero Cooler:** Air coolers are specifically designed to cool hydrocarbon products and other liquids in industrial processes.

In Super Petrochemical Ltd. (SPL), an aero cooler is used to cool hot process fluids by transferring heat to the ambient air, eliminating the need for water. It works by passing the fluid through finned tubes while fans blow air across them, effectively reducing the fluid temperature. Aero coolers help maintain process stability, enhance energy efficiency, and are especially valuable in areas with limited water resources. Their air-based operation also makes them environmentally friendly by minimizing water usage and thermal discharge. In SPL forced type Aero cooler are used.

**Motor is directly
connected to the fan
shaft**

Motor



**Hot oil
tube**

Figure 2.6: Aero Cooler

- **Agitator:** In the petrochemical industry, an agitator is a mechanical device used to mix fluids, enhance chemical reactions, and maintain uniform temperature and composition within reactors or storage tanks. It ensures proper blending of raw materials, suspends solid particles in slurries, and improves heat and mass transfer by continuously stirring the contents. Agitators also help disperse gases into liquids, which is essential in processes like oxidation or polymerization. Overall, they play a crucial role in achieving consistent product quality and efficient process performance.

2. Utility section:

In the petrochemical industry, utility equipment plays a vital supporting role in ensuring smooth, safe, and efficient plant operations. These utilities provide essential services like power, steam, air, water, and cooling.

In SPL utility equipment:

- **Boiler**
 - **Heat Exchanger**
 - **Pump**
 - **Compressor**
 - **N2 Generator**
-
- **Boiler:** A boiler is a closed vessel in which fluid (generally water) is heated. In the petrochemical industry, a boiler plays a crucial role by generating high-pressure steam used for various process applications, including heating, driving turbines, and facilitating chemical reactions. The steam produced helps maintain the required temperature and pressure conditions in reactors and distillation columns, ensuring efficient operation and energy transfer. Additionally, boilers support utility systems like water treatment, condensate recovery, and plant sanitation, making them essential for overall plant functionality and productivity.

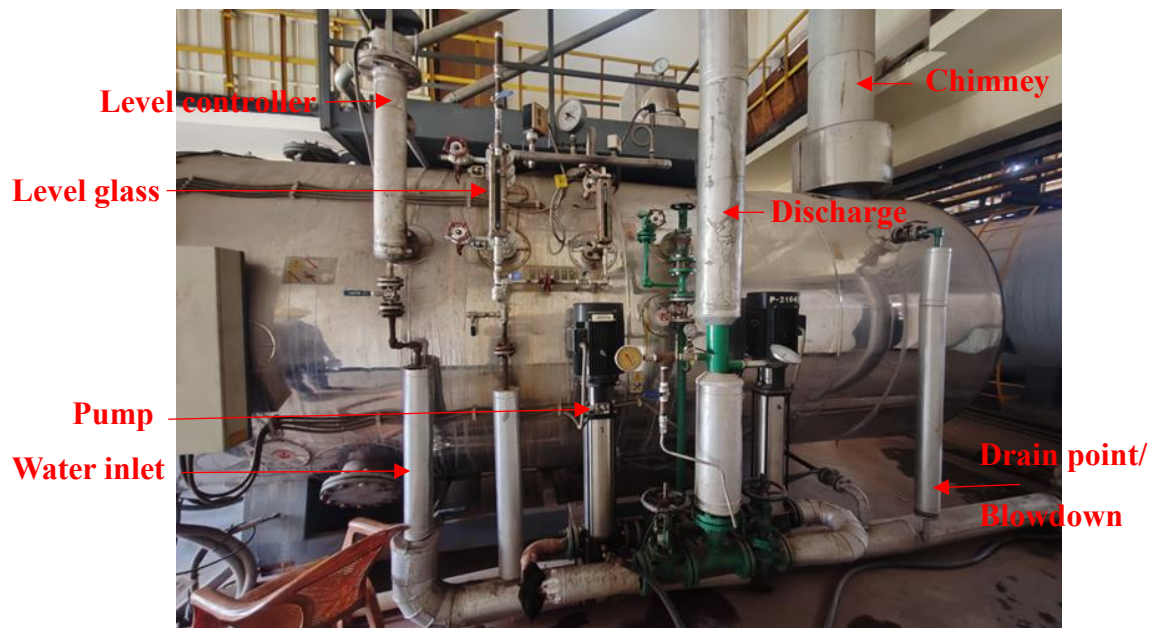


Figure 2.7: Boiler

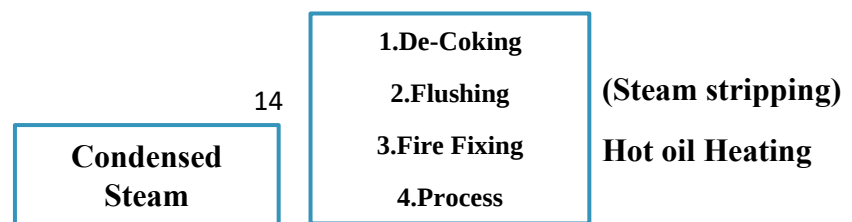
- Parameter of this Boiler

Model : ATTSU -3000-12(5-8)

Capacity: 3 tons/hr.

Max. pressure: 5-8 bar

Designed pressure: 12 bar





Deaerator

Figure 2.8: Flow Diagram of Boiler in SPL

- **Heat Exchanger:** A heat exchanger is a system used to transfer heat between a source and a working fluid. In the petrochemical industry, a heat exchanger is a critical component used to transfer heat between two fluids without mixing them. Its main function is to either heat up or cool down process streams to the required temperatures for efficient chemical reactions, separation processes, or energy recovery. Heat exchangers help maintain thermal balance within the system, improve energy efficiency by reusing heat from hot streams, and protect equipment by controlling temperatures. They are widely used in processes like distillation, reforming, and condensation, making them essential for safe and economical plant operations.

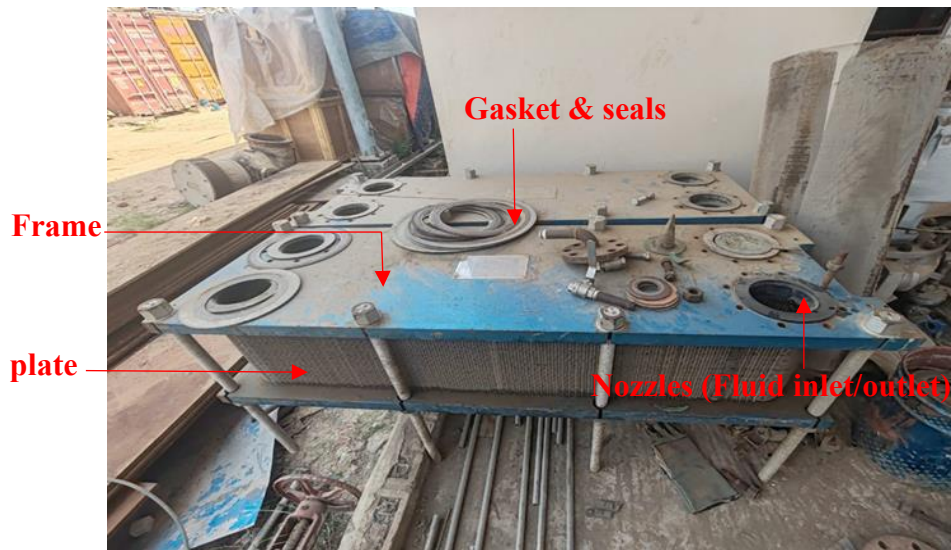


Figure 2.9: Plate & Frame Heat Exchanger

Heat exchanger type : Plate & Frame

Max. Working pressure: 10 bar

Max. Temperature : 150 C

Max. Temp. of service : 150 C



Figure 2.10: Shell & Tube Heat Exchanger

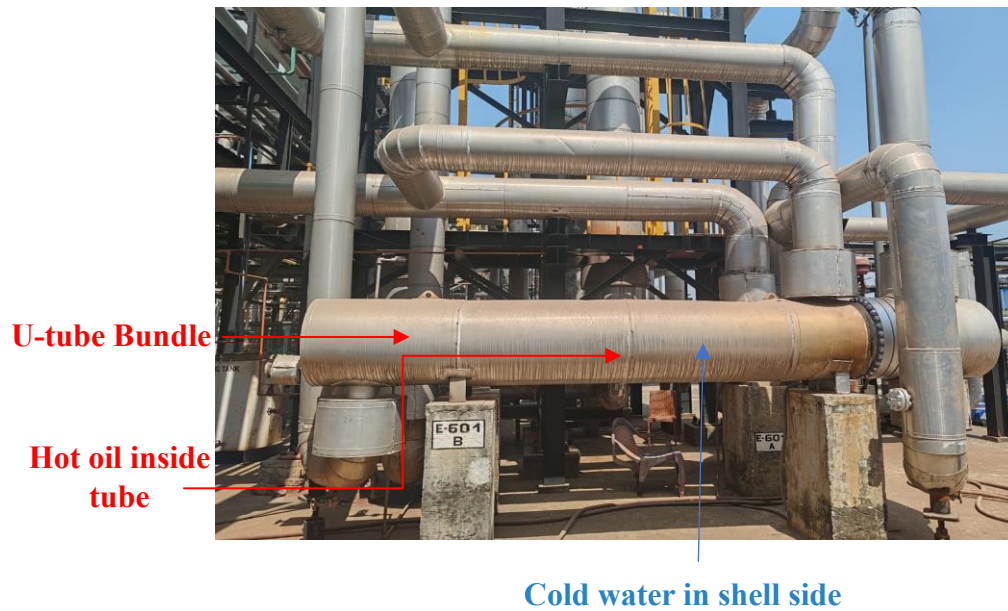


Figure 2.11: Shell & Tube Heat Exchanger

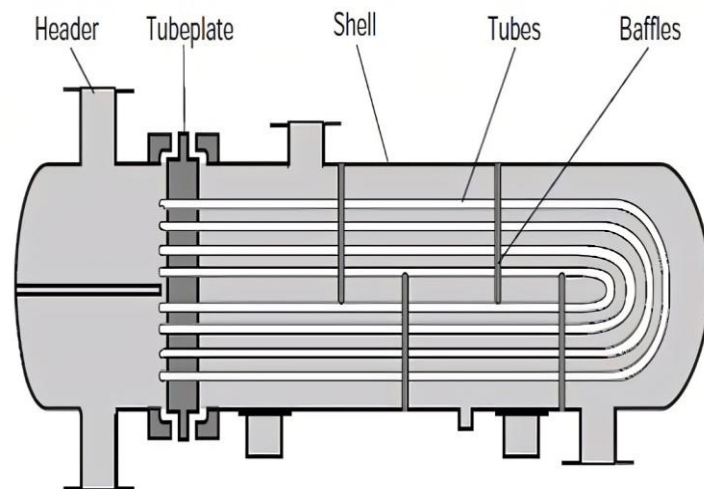


Figure 2.12: Cross-Sectional View of Shell & Tube Heat Exchanger

Chapter 3

Rotating Equipment & Static Maintenance

3.1 INTRODUCTION

Rotating equipment, such as pumps, compressors, and turbines, is essential in the petrochemical industry for fluid handling and energy transfer. Efficient operation and maintenance of this equipment are key to ensuring smooth production and minimizing downtime. Supporting this, the inventory store and distribution system manages the availability and delivery of spare parts, tools, and materials needed for maintenance and operations. Together, these systems play a vital role in maintaining plant reliability and efficiency.

3.2 ROTATING SECTION

The rotating section in a petrochemical plant includes equipment like pumps, compressors, turbines, blowers, and motors. These machines are essential for moving fluids, compressing gases, and generating mechanical power throughout the production process. Efficient operation of this equipment ensures smooth plant performance, energy efficiency, and reduced downtime. Regular maintenance, such as lubrication and vibration checks, is crucial for reliability and safety.

Rotating Section of SPL:

- Pump
- Compressor
- Aero Cooler
- Cooling Tower
- Agitator

3.3 ROTATING COMPONENT PUMP

A pump is mechanical device that moves fluids like liquids or gases, from one place to another.

Pumps are one of the most essential mechanical components in the petrochemical industry, responsible for maintaining the flow of fluids across different units in a plant. Their function goes beyond simple fluid transfer—they play a crucial role in the overall efficiency, safety, and productivity of petrochemical processes.

1. Fluid Transfer and Circulation

Pumps are used to transfer liquids such as crude oil derivatives, feedstocks (like naphtha or condensate), intermediate chemicals, and final petrochemical products between storage tanks, reactors, heat exchangers, and other processing units. This movement is necessary to keep the production cycle flowing continuously.

2. Pressure Management

In many processes, fluids must be moved against pressure gradients. Pumps generate the required pressure to overcome system resistance, ensuring the material reaches its destination efficiently. This is especially critical in high-pressure processes like catalytic reforming or distillation.

3. Feeding Process Units

Pumps feed raw materials into reactors or separators at a controlled rate. Maintaining precise flow rates is vital for achieving the right chemical reactions and ensuring product quality. Metering and dosing pumps are often used for this purpose.

4. Cooling and Heating Circuits

Pumps are also used to circulate water, steam, or other thermal fluids through heat exchangers and reactors for temperature control. This helps in maintaining process conditions and protecting sensitive equipment from thermal damage.

5. Product Loading and Unloading

Finished petrochemical products are pumped into storage tanks or transportation units like tankers and pipelines. Efficient pump operation in these stages is critical to meet delivery schedules and prevent spillage or contamination.

6. Utility Support

Pumps are used in various utility systems such as fire water systems, boiler feed systems, and chemical dosing systems, all of which are crucial for safe and uninterrupted operations.

7. Safety and Reliability

In many emergency scenarios, pumps play a life-saving role. For instance, firewater pumps ensure the availability of water during fire incidents. Reliable pump operation is therefore directly linked to plant safety.

Pump used in SPL

- Centrifugal Pump
- Reciprocating Pump
- Diaphragm Pump
- Gear Pump



Figure 3.1: centrifugal pump (Cooling Tower Section)



Figure 3.2: Diesel Pump (Boiler Section)

3.4 ROTATING COMPONENT COMPRESSOR

A compressor is a mechanical device that increases the pressure of a gas by reducing its volume. Compressors play a crucial role in the petrochemical industry by increasing the pressure of gases to facilitate their movement, storage, or processing within the plant. Their primary function is to compress and transport gases such as natural gas, hydrogen, nitrogen, air, and various process gases used in chemical reactions.

Compressors are used to:

- Feed gases into reactors at the required pressure for catalytic or thermal reactions.
- Recycle gases within closed-loop systems, maintaining efficiency and reducing waste.
- Boost gas pressure for long-distance pipeline transportation or to meet the pressure needs of downstream equipment.
- Support utility systems, such as instrument air and plant air, which are essential for automation and operational safety.

By maintaining consistent gas flow and pressure, compressors help ensure process stability, energy efficiency, and optimal production output in petrochemical operations. Proper maintenance of compressors is also vital to prevent leaks, energy loss, and equipment failure.

In SPL two types of Compressor are used.

1. Reciprocating H2 Compressor
2. Air Compressor
 - Reciprocating H2 Compressor



**Piston reciprocates
inside**

Figure 3.3: H2 Compressor



Suction pipe

Discharge pipe

Figure 3.4: Reciprocating H2 Compressor

- Air Compressor



Figure 3.5: Air Compressor



Figure 3.6: N2 Generator

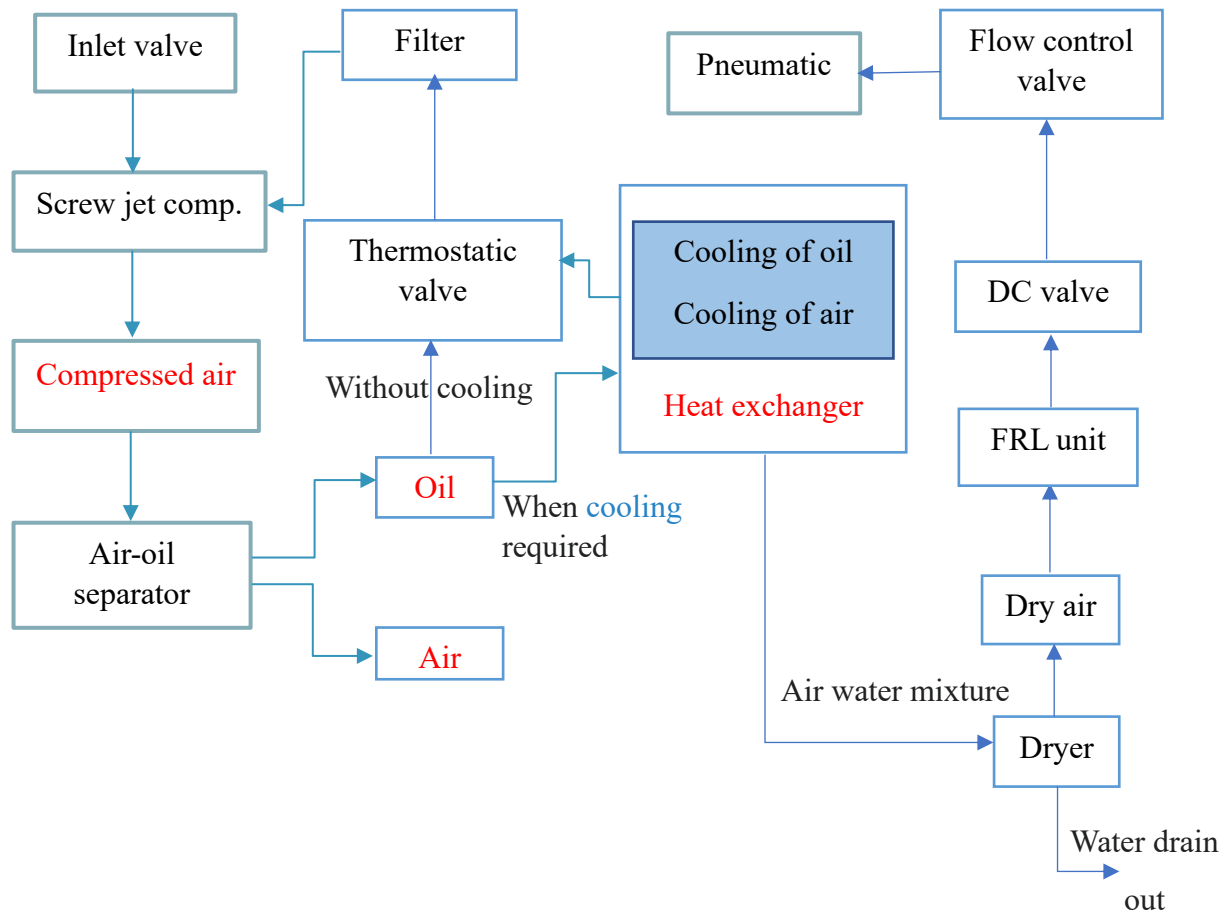


Figure 3.7: Block Diagram of Screw Jet Compressor

Copco Air Compressor

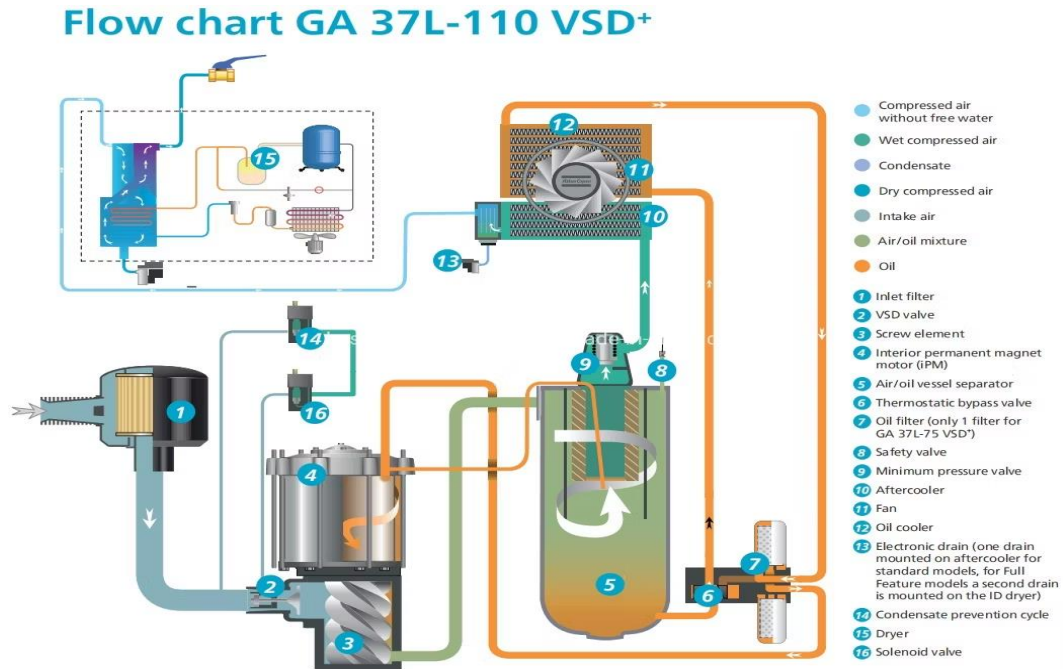


Figure 3.8: Atlas Screw Jet Compressor

3.5 STATIC MAINTENANCE

Static maintenance in the petrochemical industry involves the inspection, repair, and upkeep of non-moving (stationary) equipment to ensure safe, reliable, and efficient plant operations. This includes maintaining structural components and fixed process equipment such as pressure vessels, heat exchangers, storage tanks, piping systems, columns, boilers, and valves.

The goal of static maintenance is to prevent material degradation, corrosion, leakage, or mechanical failure that can lead to costly downtimes or safety hazards. Activities typically include cleaning, surface treatment, welding, thickness measurement, leak detection, corrosion control, pressure testing, and component replacement.

Although static equipment does not move, its proper maintenance is critical to the integrity and performance of the overall system, especially in high-pressure and high-temperature environments common in petrochemical plants.

Bearing: Bearings are essential components in the petrochemical industry, used to support and guide the rotating parts of machines like pumps, compressors, and motors. They reduce friction, ensure smooth rotation, and handle high loads under demanding conditions. Proper maintenance of bearings—such as lubrication, alignment, and inspection—is crucial for reliable equipment performance and to avoid breakdowns. Although part of rotating systems, bearing upkeep is considered a static maintenance activity within rotating equipment support.

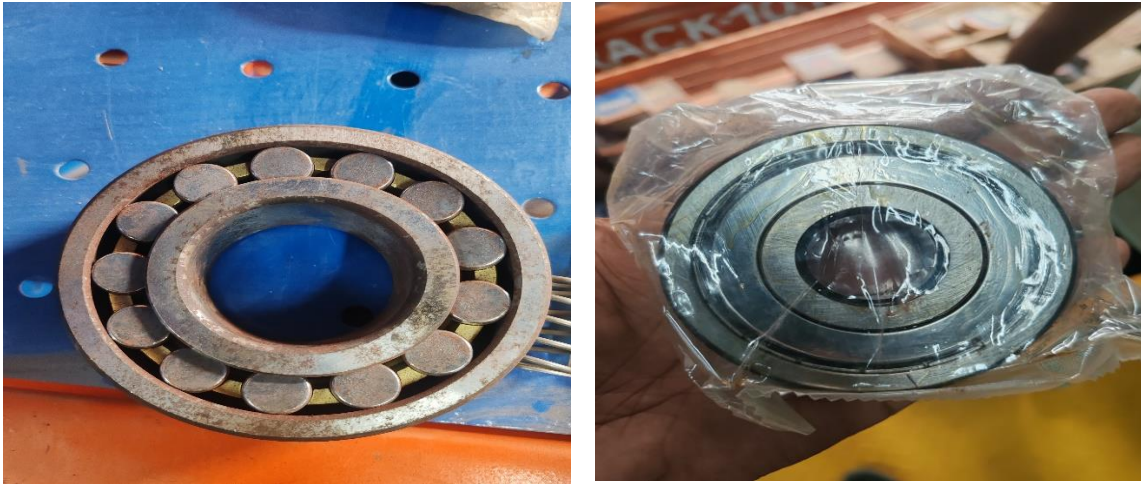


Figure 3.9: Roller Bearing (Both sided Metal sealed)

Pressure Vessel: In the petrochemical industry, pressure vessels are static equipment designed to hold gases or liquids at high pressure. They are critical for processes involving chemical reactions, storage, or separation under controlled pressure and temperature. Their integrity is vital for safety and operational stability.



Figure 3.10: Pressure Vessels

Pipe & Fittings: The pipe and fitting system in the petrochemical industry ensures the safe and efficient flow of fluids under high pressure and temperature. Pipes transport raw materials and products, while fittings—like valves, elbows, and flanges—control direction, flow, and connectivity. Made from durable, corrosion-resistant materials, these systems are critical for reliable operations, safety, and minimal downtime. Proper design, installation, and maintenance are essential for preventing leaks and supporting continuous plant performance.

Pipe & Fittings Material used in SPL Materials of pipe used in SPL is CS(ASTM-A106).



Figure 3.11: Flange & Reducer

Chapter 4

Conclusion

The 21-day industrial attachment at Super Petrochemical Limited (SPL) was a significant learning experience for me as a mechanical engineering student. Building on our prior theoretical and laboratory work, this program provided direct exposure to real-world industrial operations. I had the opportunity to observe and understand critical areas such as the utility section, rotating and static equipment maintenance, workshop practices, and mechanical tools like the bending machine. These experiences not only enhanced my technical knowledge but also improved my practical skills, especially in equipment handling, maintenance procedures, and system operations. This attachment has effectively bridged the gap between academic concepts and industrial applications, reinforcing my readiness for a professional mechanical engineering role in the petrochemical sector.

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